

Table 25. Embedded reset and power control block characteristics (continued)

Symbol	Parameter	Conditions	Min	Typ	Max	Unit
$V_{PDR}^{(1)}$	Power-down reset threshold	-	1.88	1.92	1.96	V
V_{BOR1}	Brownout reset threshold 1	V_{DD} rising	2.05	2.10	2.18	V
		V_{DD} falling	1.95	2.00	2.08	
V_{BOR2}	Brownout reset threshold 2	V_{DD} rising	2.20	2.31	2.38	V
		V_{DD} falling	2.10	2.21	2.28	
V_{BOR3}	Brownout reset threshold 3	V_{DD} rising	2.50	2.62	2.68	V
		V_{DD} falling	2.40	2.52	2.58	
V_{BOR4}	Brownout reset threshold 4	V_{DD} rising	2.80	2.91	3.00	V
		V_{DD} falling	2.70	2.81	2.90	
$V_{hyst_POR_PDR}$	Hysteresis of V_{POR} and V_{PDR}	-	-	20	-	mV
V_{hyst_BOR}	Hysteresis of V_{BORx}	-	-	100	-	mV
$I_{DD(BOR)}^{(1)}$	BOR consumption from V_{DD}	-	-	2.2	2.5	μA

1. Specified by design. Not tested in production.

5.3.4 Embedded voltage reference

The parameters given in [Table 26](#) are derived from tests performed under the ambient temperature and supply voltage conditions summarized in [Table 23: General operating conditions](#).

Table 26. Embedded internal voltage reference

Symbol	Parameter	Conditions	Min	Typ	Max	Unit
V_{REFINT}	Internal reference voltage	-	1.182	1.212	1.232	V
$t_{S_vrefint}^{(1)(2)}$	ADC sampling time when reading the internal reference voltage	-	4	-	-	μs
$t_{start_vrefint}^{(2)}$	Start time of reference voltage buffer when ADC is enable	-	-	8	12	μs
$I_{DD(VREFINTBUF)}^{(2)}$	V_{REFINT} buffer consumption from V_{DD} when converted by ADC	-	9	13.5	23	μA
$\Delta V_{REFINT}^{(2)}$	Internal reference voltage spread over the temperature range	$V_{DD} = 3 V$	-	30	50	mV
T_{Coeff}	Average temperature coefficient	-	-	20	70	ppm/°C
A_{Coeff}	Long term stability	1000 hours, $T = 25\text{ }^{\circ}C$	-	300	1000	ppm
$V_{DDCoeff}$	Voltage coefficient	$3.0 V < V_{DD} < 3.6 V$	-	250	1200	ppm/V

1. The shortest sampling time can be determined in the application by multiple iterations.

2. Specified by design. Not tested in production.